

Table 1

Title (has dataset)	WANDS : Dataset for Product Search Relevance
Harvard reference	Chen, Y., Liu, S., Liu, Z., Sun, W., Baltrunas, L. and Schroeder, B., 2022. WANDS: Dataset for Product Search Relevance Assessment. In: M. Hagen, V. Verberne, C. Macdonald, C. Seifert, K. Balog, K. Nørvåg and V. Setty, eds. <i>Advances in Information Retrieval: 44th European Conference on IR Research, ECIR 2022, Stavanger, Norway, April 10–14, 2022, Proceedings, Part II</i> . Cham: Springer, pp. 128–141. Available at: https://doi.org/10.1007/978-3-030-99736-6_9 [Accessed 18 September 2026]
Link	https://github.com/wayfair/WANDS/tree/main https://arxiv.org/pdf/2206.06588v1
Modelling	BM25 and nDCG@10
Preprocessing techniques	No preprocessing of data mentioned in the article
Datasets used	Wayfair a e-commerce and home goods company Annotation Dataset (WANDSA) a human labelled ecommerce search relevance dataset.
Key takeaways	WANDS label dataframe uses labels like exact , partial and irrelevant to score the relationship

	between customer search query and product relevance WANDS product listening contains metadata that aids in relevance ranking , the BM25 model is also able to use product meta data to guide query and product pairing matches which helps the ranking And lastly Customer behaviour according to the researchers is not sufficient to gauge product relevance , due to product placements therefore the use of human intervention to label exact or partial or relevant is crucial .
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Table 2

Title	Shopping Queries Dataset: A Large-Scale ESCI Benchmark for Improving Product Search Amazon
Harvard reference	Reddy, C.K., Màrquez, L., Valero, F., Rao, N., Zaragoza, H., Bandyopadhyay, S., Biswas, A., Xing, A. and Subbian, K. (2022) <i>Shopping Queries Dataset: A Large-Scale ESCI Benchmark for Improving Product Search</i> . arXiv:2206.06588. Available at: https://arxiv.org/abs/2206.06588 .
Link	Available at: https://arxiv.org/abs/2206.06588 .
Modelling	BM25 Mpnet as a text-based supervised model to translate text from Spanish and Japanese
Preprocessing techniques	Based on the ESCI criteria as defined in the article , based on the query made by the customer the product match using a binary task

	to encode of 1.0 or 0.0 this than ranks the product relevance
Datasets used	Amazon Shopping Queries Dataset (ESCI dataset). With 48,300 unique queries and 1,118,117 query–product judgement rows.
Key takeaways	human annotation was needed to decide on the relevance score of the search result , human judgment using ESCI this leads to a reliable judgement when training models Relevance label and score was either 1.0 for exact or 1.1 or not accurate (S/C/I) this was binary training target Lastly two models (BM25 vs neural Cross Encoder/MPNet) approaches of of rating search relevance was used , BM25 which measured key words/terms matching in the the customer search query and the product title description match than (Lexical approach) the second model was neural Cross, encoder/MPnet which measures if the semantic meaning of the search terms matches nDCG measures weather the correct products are ranked at the top and the article found that Neural Cross encoder achieved a nDCG score, however researchers caution that BM25 lower result may be due to poor Japanese non-specific preprocessing.

Table 3

Title	Exploring the Viability of Synthetic Query Generation for Relevance Prediction
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(Has dataset)	
Harvard reference	A valid Harvard reference for the publication
Link	https://arxiv.org/pdf/2305.11944
Modelling	MT5 encoder , which is a classification model that is able to read the query , product , product description and then predicts the relevance class in the ESCI database . The second model is RankT5 which ranks models by giving query and product pairs relevance scores which are ranked
Preprocessing techniques	all the four datasets had different preparation processes , the MS-MARCO dataset had a binary label of 0=irrelevant and 1 =relevant , The second data set the Amazon ESCI , the researchers kept all the four labels and then transformed each example into an input with relevance label , product title , product description and the customer query and the output , the Wands dataset was a target so in the preparation the outputs were exact , partial and irrelevant labels , furthermore the dataset was not split or trained instead all 233 , 448 examples were evaluation set , lastly Home Depot Dataset 279 of the examples had software parsing issue there for only 73 789 examples were used . From all of the datasets above all duplicates were removed , and the datasets were split 90:10
Datasets used	The article makes use of different data sets , two are from training and the other two are target and evaluation dataset . The training dataset used are Amazon ESCI and MS- Marco and the target

	evaluation data sets are WANDS and Home Depot
Key takeaways	ESCI and MS -Marco are source dataset that will be used to train WAnds and Home depot are target datasets the researchers aim is to test if relevance can be transferred to a new ecommerce dataset so according to the researchers the ESCI dataset performed better in the training even though the MARCO dataset was bigger, this was due to the Marco dataset being more general and the ESCI dataset being ecommerce specific to Amazon . Lastly The RankT5 model pair query and products using relevance which i believe is the exact problem my problem statement aims to solve and model

Table 4 : Dataset used for modeling

Name	WANDS, Dataset for Product Search Relevance
Harvard reference	Chen, Y., Liu, S., Liu, Z., Sun, W., Baltrunas, L. and Schroeder, B., 2022. WANDS: Dataset for Product Search Relevance Assessment. In: M. Hagen, V. Verberne, C. Macdonald, C. Seifert, K. Balog, K. Nørvgå and V. Setty, eds. <i>Advances in Information Retrieval: 44th European Conference on IR Research, ECIR 2022, Stavanger, Norway, April 10–14, 2022, Proceedings, Part II</i>. Cham: Springer, pp. 128–141. Available at: https://doi.org/10.1007/978-3-030-99736-6_9
Link	https://www.kaggle.com/datasets/deeguy/wayfair-wands-dataset?select=query.csv

Number of instances	42 994 Products 480 Query from customers 233,448 labelled query–product pairs. No test or training split mentioned or specified
Number of feature columns	3 Query.csv Product.csv Label.csv
Number of classes (if applicable)	3 Relevance Classes Exact , products that match customer search query exactly Partial – products that are partially relevant to search query Irrelevant , products that are not relevant to search query at all.
Target range (if applicable)	N/A Target
Origin	Wayfair a e-commerce and home goods company Annotation Dataset (WANDSA) a human labelled ecommerce search relevance dataset.